

Unnamed_Unit

Unit 2 - Bringing Ideas to Life

PDF Documents

PDF Document 1: 1738213122719-Chapter 5 - Unit 2 - Bringing Ideas to Life.pdf

This PDF document is included in the unit materials.



UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

WHY BUILD A WINDOW CLEANING ROBOT?

I sometimes had fun washing windows and shaking out blankets with my dad. However, because of our busy lives, it is not easy to enjoy cleaning. In times like this, if a robot takes over the cleaning, wouldn't it be possible to have fun with your dad doing something different?



This year was a good harvest, so there were a lot of crops. My mother and grandmother feel uncomfortable because they have to work continuously on hot days without being able to rest. I want my mom and grandma to rest comfortably in the shade and let the intelligent robot farmer help me with the harvest. How should I prepare to create the world I envisioned



In today's world, technology is everywhere and changing how we live, work, and solve problems. Let's see how angles, sensors, and movement can be used to control a robot's motion.



ABOUT MOTOR A MODULE

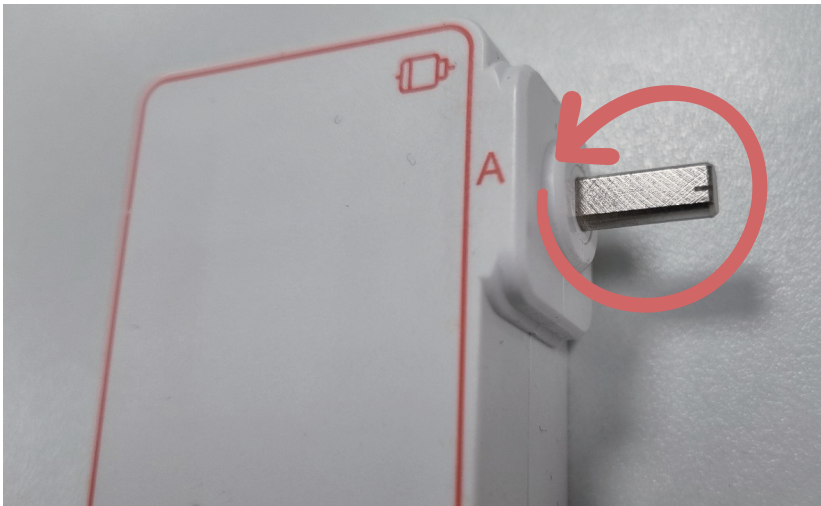
set motor A 1 ▼ angle to 0 °

set motor A 1 ▼ angle to 0 °, speed to 100 %

change motor A 1 ▼ angle to clockwise ▼ 0 °

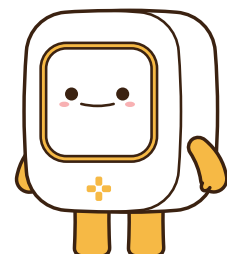
Motor A (same as B) can control speed and angle, and for angles, there is a block that allows you to specify the angle, the speed at which it moves by the angle, and the direction of rotation of the angle.

CODING ACCORDING TO ANGLES



TIPS

- The grooved side of the motor shaft is the 0 point.
- Used to determine location when inserting the horn

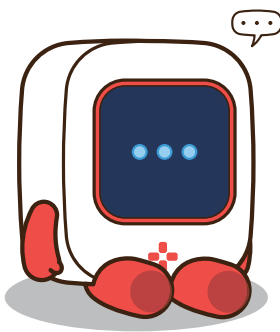
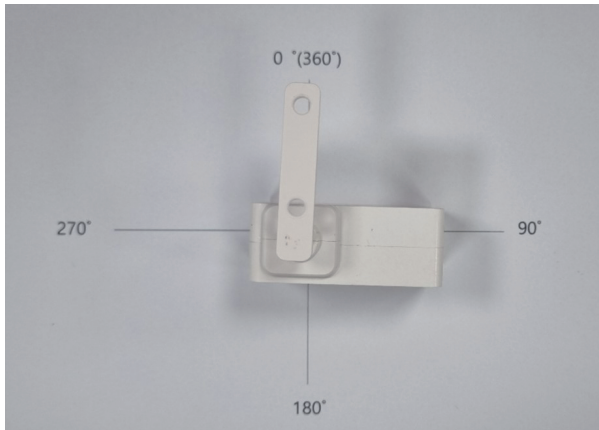


TRANSLATE THE CODING EXAMPLE FOR MOVING TO A POSITION APPROPRIATE FOR THE ANGLE.

Let's place the motor with the horn, as shown in the picture, and code it.

(Not pictured, but coding by connecting the network module)

When you run the code, it checks whether the horn points to the position according to the set angle.



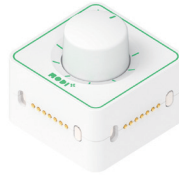


Required Modules & Brainstorming

Based on the story, which MODI modules do you need for this creation?



BUTTON



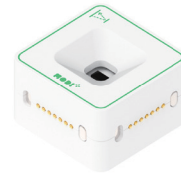
DIAL



ENVIRONMENT



JOYSTICK



TOF



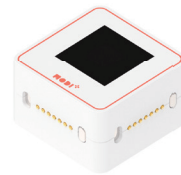
SPEAKER



IMU



LED



DISPLAY

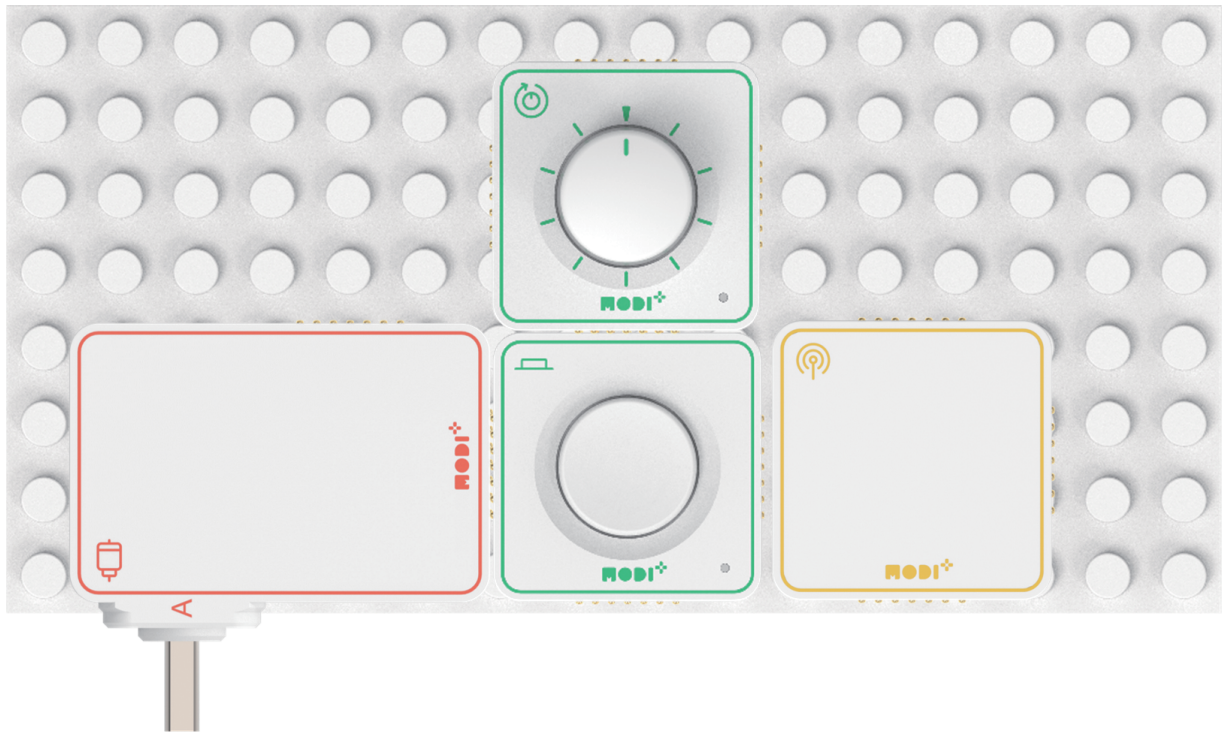


MOTOR A

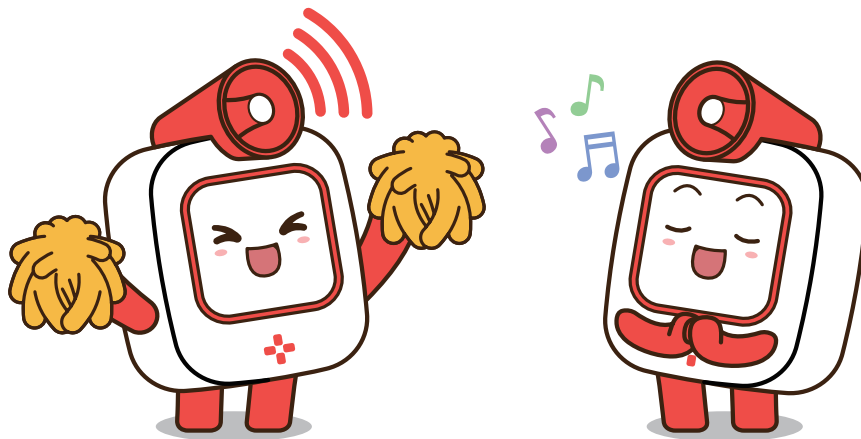


MOTOR B

HOW DOES THE CREATION LOOK LIKE?



This translation focuses on monitoring the cleaning range, determining the angle, and coding it accordingly.





Module Functionality

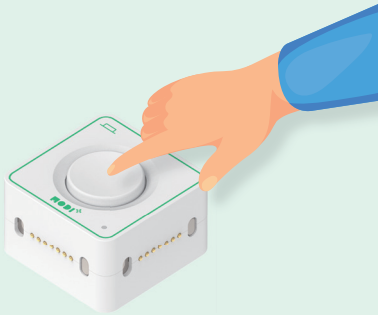
Let's look detail about the Button module.



- ✓ clicked
- double clicked
- pressed
- toggle on

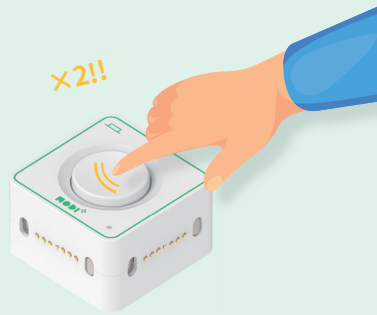
Button can receive 4 types of information.

1. Click



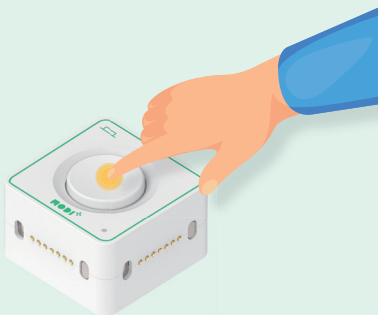
Detects if Button is clicked or not.
Range: True(Click once) : 100 / False : 0

2. Double click



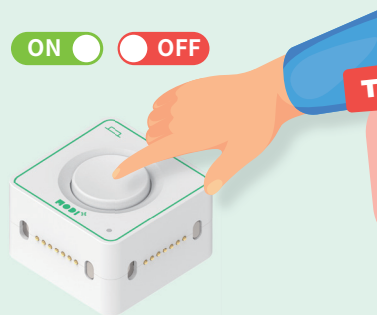
Detects if Button is double - clicked or not.
Range: True(Click twice) : 100 / False : 0

3. Press



Show the press status of Button.
Range: True(Pressed) : 100 / False : 0

4. Toggle



Changes on / off when Button is clicked.
Range: True(On) : 100 / False : 0

THINK ABOUT IT!

Do you remember which action Button can receive?

